

MASAFUMI ENDO

✉ masafumi.endo@ieee.org | 🌐 masafumiendo.github.io

🎓 Google Scholar | 🐙 GitHub | 🔗 LinkedIn

📍 Shibuya Scramble Square, 2-24-12 Shibuya, Shibuya-ku, Tokyo 150-6121, Japan

Research Interests

Embodied AI for everyday scenarios, focusing on decision-making and scene understanding.

Keywords: planning under uncertainty, scene understanding, integrated planning and learning, foundation models for robotics, planetary exploration

Education

Keio University, Greater Tokyo Area, Japan

Ph.D. in Engineering

Sep. 2021 – Mar. 2025

Oregon State University, Oregon, The United States

M.S. in Robotics, GPA: 3.82/4.00

Sep. 2019 – Aug. 2021

Tohoku University, Sendai, Japan

M.Eng. in Aerospace Engineering, GPA: 4.00/4.00

Apr. 2017 – Mar. 2019

Kobe University, Kobe, Japan

B.Eng. in Mechanical Engineering

Apr. 2013 – Mar. 2017

Research Experience

CyberAgent AI Lab – Activity Understanding Team, Japan

Apr. 2025 – Present

Research Scientist

- Working on embodied AI for everyday scenarios, focusing on decision-making and scene understanding.
- Developed planning methods for service robots in human environments (three arXiv preprints, 2026).

NASA Jet Propulsion Laboratory – Robotic Mobility Group, The United States

Aug. 2024 – Nov. 2024

Research Intern, advised by Dr. Masahiro Ono, Dr. Ashkan Jasour, and Mr. Rohan Thakker

- Worked on establishing a new TAMP method for resilient autonomy under uncertainty (IEEE ICRA, 2025).

Field AI, The United States

Oct. 2023 – Dec. 2023

Research Intern, advised by Dr. David D. Fan

- Worked on establishing reliable ML-based robotic perception in challenging environments.

OMRON SINIC X Corporation, Japan

Jun. 2022 – Mar. 2023

Research Intern, advised by Dr. Ryo Yonetani and Dr. Tatsunori Taniai

- Developed an end-to-end probabilistic ML model for adaptive, uncertainty-aware planetary rover navigation (Scientific Reports, 2026).
- Proposed a path planning method robust against erroneous traversability prediction via ML models, representing multimodal slip distribution for heterogeneous deformable terrains (IEEE ICRA, 2023).

Keio University – Space Robotics Group, Japan

Sep. 2021 – Mar. 2025

JSPS DC1 Research Fellow, advised by Prof. Genya Ishigami

- Mentoring a graduate student to work on model predictive path integral control techniques with Dr. Kohei Honda from Nagoya University (IEEE CASE, 2025 & IEEE/SICE SII, 2025).
- Developed a Pytorch-based simulation platform for benchmarking off-road navigation algorithms with built-in ML models for probabilistic traversability prediction (IEEE ICRA WS, 2024).
- Proposed a sampling-based informative path planner with probabilistic risk inference to actively learn latent traversability for planetary rovers (IEEE RA-L, 2022).

Graduate Research Assistant, advised by Prof. Burak Sencer

- Developed a cycle-time prediction method in combination with an analytical dynamics model and ML architecture for CNC machines (Int. Manuf. Sci. Eng. Conf., 2021 & CIRP Annals, 2022).

Tohoku University – Space Robotics Laboratory, Japan

Apr. 2017 – Mar. 2019

Master Student, Advised by Prof. Kazuya Yoshida, Prof. Kenji Nagaoka, and Prof. Toshinori Kuwahara

- Presented a vision-based approach with machine learning for predicting wheel slip risk (Robotica, 2021).
- Developed a vision-based traveling state estimation method for rovers (JSME Conf. Robot. Mechatron., 2019).
- Implemented a micro-satellite trajectory estimation method (IAA NE Asia Symp. Small Satell., 2017).

Kobe University – Laboratory of Energy Conversion Engineering, Japan

Apr. 2016 – Mar. 2017

Undergraduate Student, Advised by Prof. Tsuyoshi Kawanami

- Built a soft capsule containing PCM to verify its ability to keep thermal energy.

Publications

Preprints

1. ReVNM: Learning-Based Visual Navigation from a Remote Camera
M. Eguchi, K. Honda, **ME**, Y. Yoshimura, and R. Yonetani, *arXiv preprint*, arXiv:2609.28976, 2026.
2. Commonsense-Grounded Path Planning from Abstract Instructions
ME, K. Honda, and R. Yonetani, *arXiv preprint*, arXiv:2609.22813, 2026.
3. Orienteering Problem with Uncertain Time-Varying Rewards: Framework and Benchmark for Everyday Service Robotics
ME, K. Honda, Y. Jinnai, and R. Yonetani, *arXiv preprint*, arXiv:2608.18672, 2026.

Journal Papers

4. Deep Probabilistic Traversability with Test-time Adaptation for Uncertainty-aware Planetary Rover Navigation
ME, T. Taniai, and G. Ishigami, *Scientific Reports*, 2026.
5. Active Traversability Learning via Risk-aware Information Gathering for Planetary Exploration Rovers
ME and G. Ishigami, *IEEE Robotics and Automation Letters*, vol. 7(4), pp. 11855-11862, 2022.
6. Accurate Prediction of Machining Cycle Times by Data-driven Modelling of NC System's Interpolation Dynamics
ME and B. Sencer, *CIRP Annals*, vol. 71(1), pp. 405-408, 2022.
7. Terrain-Dependent Slip Risk Prediction for Planetary Exploration Rovers
ME, S. Endo, K. Nagaoka, and K. Yoshida, *Robotica*, vol. 39(10), pp. 1883-1896, 2021.

Conference Proceedings

8. DRPA-MPPI: Dynamic Repulsive Potential Augmented MPPI for Reactive Navigation in Unstructured Environments
T. Fuke, **ME**, K. Honda, and G. Ishigami, *IEEE International Conference on Automation Science and Engineering*, 2025.
9. Risk-aware Integrated Task and Motion Planning for Versatile Snake Robots under Localization Failures
A. Jasour*, G. Daddi*, **ME***, T. S. Vaquero*, M. Paton, M. P. Strub, S. Corpino, M. Ingham, M. Ono, R. Thakker (*Equal contributions), *IEEE International Conference on Robotics and Automation*, 2025.
10. Towards Local Minima-free Robotic Navigation: Model Predictive Path Integral Control via Repulsive Potential Augmentation
T. Fuke, **ME**, K. Honda, and G. Ishigami, *IEEE/SICE International Symposium on System Integration*, 2025.
11. BenchNav: Simulation Platform for Benchmarking Off-road Navigation Algorithms with Probabilistic Traversability
ME, K. Honda, and G. Ishigami, *IEEE ICRA 2024 Workshop on Resilient Off-road Autonomy*, 2024.

12. Risk-aware Path Planning via Probabilistic Fusion of Traversability Prediction for Planetary Rovers on Heterogeneous Terrains
ME, T. Tani, R. Yonetani, and G. Ishigami, *IEEE International Conference on Robotics and Automation*, pp. 11852-11858, 2023.
13. Traveling State Estimation of a Wheeled Robot using ToF Camera and Torque Sensor for Lunar/Planetary Exploration
S. Endo, ME, K. Nagaoka, and K. Yoshida, *JSME Conference on Robotics and Mechatronics*, 2P1-T03, 2019.
14. Quality of the 3D Point Cloud of a Time-of-Flight Camera Under Lunar Surface Illumination Conditions: Impact and Improvement Techniques
K. Uno, L.-J. Burtz, ME, K. Nagaoka, and K. Yoshida, *International Symposium on Artificial Intelligence, Robotics and Automation in Space*, #10a-2, 2018.

Conference Presentations

15. Active Traversability Learning via Risk-aware Information Gathering for Planetary Exploration Rovers
ME and G. Ishigami, *IEEE/RSJ International Conference on Intelligent Robots and Systems*, 2022.
16. Machining Cycle-Time Prediction by Machine Learning of CNC Interpolator Dynamics
ME and B. Sencer, *International Manufacturing Science and Engineering Conference*, 2021.
17. Microsatellite Bus System Technologies of Tohoku University
T. Kuwahara, Y. Sakamoto, S. Fujita, Y. Sato, R. Taba, H. Katagiri, ME, P. Tangdhanakarn, T. Honda, and K. Yoshida, *IAA North East Asia Symposium on Small Satellites*, 2017.

Fellowships/Scholarships/Awards

Fellowships

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| JSPS Overseas Challenge Program for Young Researchers | Aug. 2024 – Nov. 2024 |
| Sponsored by Japan Society for the Promotion of Science (1,400,000 yen) | |
| JSPS Research Fellowship for Young Scientists DC1 | Apr. 2022 – Mar. 2025 |
| Sponsored by Japan Society for the Promotion of Science (200,000 yen/month + research grant) | |

Scholarships

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| Research Encouragement Scholarship for Graduate Students | Apr. 2024 – Mar. 2025 |
| Sponsored by Keio University (700,000 yen/year) | |
| Research Encouragement Scholarship for Graduate Students | Apr. 2023 – Mar. 2024 |
| Sponsored by Keio University (500,000 yen/year) | |
| JEES/Mitsubishi Corporation Scholarship | Apr. 2023 – Aug. 2024 |
| Sponsored by JEES and Mitsubishi Corporation (100,000 yen/month) | |
| Fujiwara Scholarship | Oct. 2021 – Mar. 2022 |
| Sponsored by Keio University (500,000 yen/year) | |

Awards

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| NAMRC 49 and MSEC 2021 Student Support Award | Apr. 2021 |
| Sponsored by National Science Foundation | |

Teaching Experience

Teaching Assistant

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| Computer-Aided Design and Manufacturing (ME 413, OSU) | Spring 2020 |
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Paper Review

- IEEE International Conference on Robotics and Automation (ICRA) 2023, 2024, 2026, 2027
- IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS) 2022 – 2026

- IEEE Robotics and Automation Letters (RA-L)
- IEEE Transactions on Robotics (T-RO)
- IEEJ Transactions on Electrical and Electronic Engineering

Expertise/Pertinent Skills

Programming:

Python, C++, Julia, MATLAB

Software:

Pytorch, ROS, Git

Language:

English, Japanese (Native)